TECHNICAL SPECIFICATION

Of

ACSS Conductor Equivalent to ACSR Panther /Lynx /Zebra /Moose Conductors

(Aluminium Conductor, Steel Supported Wire-RW)

Technical Specifications

1.0 SCOPE

This specification provides for design, manufacture, engineering, inspection, stage testing and testing before dispatch, packing and delivery, unloading and stacking at destination stores/ site in Andhra Pradesh of steel cored Aluminium conductor as per IS – 398 part-2, 5 for PANTHER /LYNX /ZEBRA /MOOSE specified herein for its satisfactory operation in various 132kV /220kV lines and substations of the State.

ACSS type HTLS conductors (RW) equivalent to ACSR Panther, Lynx, Zebra and Moose shall be used as power conductors on 132kV and 220kV transmission lines of the owner. ACSS HTLS Conductor is to be used as power conductor on double and quad circuits of transmission lines of the owner.

1.0.1 CONFLICT OF STANDARDS

Equipment conforming to other internationally accepted standards, which ensure equal or higher quality than the standards mentioned above, would also be acceptable. In case the Bidders who wish to offer material conforming to the other standards, including JIS (Japan Standards association) salient points of difference between the standards adopted and the specific standards shall be clearly brought out in relevant schedule. Four copies of such standards with authentic English Translations shall be furnished along with the offer. In case of conflict the order of precedence shall be (i) IS (ii) IEC (iii) Other standards. In case of any difference between provisions of these standards and provisions of this specification, the provisions contained in this specification shall prevail.

1.1 CLIMATIC CONDITIONS:

The conductor is being installed directly in air with the support of Towers and with the help of Hardware and accessories and Insulators. The conductor shall be therefore suitable for satisfactory operation under the tropical climatic conditions listed in Clause 23 of Section-III.

1.1.1 GENERAL TECHNICAL REQUIREMENT

1.1.1.1 The ACSS conductor shall be suitable for being installed directly in air supported on suspension insulator strings or anchored through tension insulator strings at the power cross arms of single circuit, double circuit and multi circuit transmission line towers.

1.1.1.2 The conductor shall therefore be suitable for satisfactory operation under the tropical climatic conditions listed under the clause 1.4. The applicable design particulars of the conductor to be used on the lines is furnished in Clause 23 of Section-III.

1.1.2 CONDUCTOR PARAMETERS:

1.1.2.1 The Parameters of individual strands and ACSS Conductor, shall be in accordance with the values given in Clause 1.3 in this Section.

1.1.2.2 Creep in a conductor is attributed partly due to settlement of strands partly due to non-elastic elongation of metal when subjected to load. The manufacturer of conductor shall furnish the amount of creep which will take place in 10, 20, 30, 40 and 50 years along with the supporting calculations. The calculations should be based on every day temperature of 32^0C and everyday tension of 25% of UTS of conductor in case of 132kV and 220kV Tranmsission Lines. The creep calculations shall be approved by APTRANSCO before final inspection of conductor.

1.1.3 S U R F A C E CONDITIONS:

All aluminum and steel strands shall be smooth, and free from all imperfections, spills. The finished conductor shall be smooth, compact, uniform and free from all imperfections including spills and splits, die marks. Scratches, abrasions, scuff marks, kinks (protrusion of wires). Dents press marks, cut marks. Wire cross-over, over-riding looseness, pressure and/or unusual bangle noise on tapping, material inclusions, while rust, power formation or black spots (on account of reaction with trapped rain water etc.,), dirt, grit, etc. The surface of conductor shall be free from points, sharp edges, abrasions or other departures from smoothness or uniformity of surface contour that would increases radio interference and corona losses. When subject to tension up to 50% of the ultimate strength of the conductor, the surface shall not depart from the cylindrical form nor any part of the component parts or strands move relative to each other in such a way as to get out of place and disturb the longitudinal smoothness of the conductor.

1.1.4 Description of ACSS Type High Temperature Low Sag (HTLS) Conductor (equivalent to PANTHER) and its Technical Requirements

The offered ACSS Type HTLS Conductor equivalent to ACSR Panther / Lynx / Zebra / Moose conductors shall be capable of carrying the rated current corresponding to the ampacity as declared in the Guaranteed Technical Particulars (GTP).

The physical and operating performance requirements of the transmission line with HTLS conductor are mentioned below. The bidder shall offer ACSS Type HTLS conductor complying with the specified requirements. The Bidder shall indicate particulars of the proposed conductor in the relevant GTP. During detailed engineering stage, the Supplier shall submit detailed calculation in support of the various parameter quoted by him in the Guaranteed Technical Particulars and the same shall be verified to establish compliance with

the specified requirements i.e. Current Carrying Capacity (Clause 1.2), Physical dimensional (Clause 1.3), PLS Cadd cable data criteria (Clause 1.4). In case any changes are required in the design of ACSS Type HTLS conductors to comply the stipulated requirement of the technical specification, the same shall be carried out without any extra cost to Purchaser. The AC resistance/ DC resistance at temperature corresponding to the specified ampacity shall not exceed the value guaranteed by the bidder in bid proposal Sheets.

1.2. Current Carrying Capacity /Ampacity Requirements

1.2.1 The offered ACSS conductor shall be capable of carrying the rated current corresponding to the declared ampacity in the Guaranteed Technical Particulars (GTP) of 50 Hz alternating current under the ambient conditions & maximum conductor sag specified below while satisfying other specified technical requirements/ parameters:

Elevation above sea level = 0 m
Ambient temperature: 45 deg C
Solar Absorption coefficient =0.8
Solar Radiation = 1045 watt/sq.m
Emissivity Constant= 0.45
Wind velocity considering angle between wind & axis of conductor as 90 degrees = 0.56m/sec
Effective angle of incidence of sun's rays= 90 deg

The calculations for Ampacity shall be based on IEEE Standard 738-2006 in SI units. Ratio of AC resistance & DC resistance for ACSS conductor shall be calculated on the basis of *Chapter-2, EPRI AC Transmission Line Reference Book – 200 kV and Above* or the formulae indicated as follows:-

$R_{ac} = R_{dc} X (1+ 0.00519 X (mr)^n X k_1 + k_2)$

where, $mr = 0.3544938/ (R_{dc})^{½}$

if $mr < 2.8$, then $n = 4- 0.0616 + 0.0896 X mr – 0.0513 X(mr)^2$

if $mr > 2.8 < 5.0$, then $n = 4+ 0.5363 -0.2949X mr +0.0097 X(mr)^2$

$k_1 = \{\cos (90 (d/D)^P)\}^{2.35}$ where,

$p = 0.7 + 0.11Xmr – 0.04Xmr^2 + 0.0094Xmr^3$

$k_2 = 0.15$ for single aluminium layer ACSS conductor

$= 0.03$ for three aluminium layer ACSS conductor

= 0.003 for two or four aluminium layer ACSS conductor

= 0 for composite core type ACSS conductor where,

D= conductor outer diameter in metres d = conductor inner diameter in metres

R_{dc} = DC resistance of conductor at given temperature, ohms/ km

R_{ac} = AC resistance of conductor at given temperature, ohms/ km

The Supplier shall furnish calculations for the ampacity based on the above for the proposed ACSS type conductor during detailed engineering.

1.2.2 The design of the conductor shall be suitable for operation at the steady-state conductor temperature corresponding to the rated current carrying capacity of the offered ACSS conductor under the specified ambient conditions, based on ampacity calculations mentioned above. For the purpose of evaluation, the minimum ampacity corresponding to the equivalent ACSR conductor size shall be considered as follows:

- 600 A for Panther/Lynx equivalent conductors,
- 1000 A for Zebra equivalent conductor,
- 1262A for Moose equivalent conductor.

The bidder shall indicate the maximum permissible conductor temperature for continuous operation without any deterioration of its electrical, mechanical and metallurgical properties. The maximum permissible conductor temperature for continuous operation shall not be less than 180°C. The bidder shall also furnish the maximum permissible conductor temperature for short-term operation along with the permissible duration of such short-term operation.

The Ultimate Tensile Strength (UTS) of the conductor at ambient temperature and at the maximum continuous operating temperature shall be declared in the GTP. Further, UTS of conductor achieved at elevated temperature (i.e. at the designed maximum Continuous conductor operating temperature) shall be not less than 80% of UTS at ambient temperature declared in the GTP.

1.3 Technical Particulars of ACSS Conductor

The ACSS conductor shall meet the following technical requirements:

Sl. No	Description	Requirements for			
		ACSS LYNX	ACSS PANTHER	ACSS ZEBRA	ACSS MOOSE
1	Diameter in mm	≤ 19.53	≤ 21.00	≤ 28.62	≤ 31.77
2	Weight of conductor*** in kg/kM	≤ 842	≤ 974	≤ 1621	≤ 2004
3	Minimum Ultimate Tensile strength** in kN	≥ 79.80	≥ 89.67	≥ 130.31	≥ 161.20
4	Sectional Aluminium area in sq.mm	≤ 183.41	≤ 212.10	≤ 428.90	≤ 528.5
5	Total Sectional area of Conductor in sq.mm	≤ 226.21	≤ 261.60	≤ 484.48	≤ 597
6	Required Ampacity* in Amperes	≥ 600	≥ 600	≥ 1000	≥ 1262
7	Maximum sag at above Ampacity in m	≤ 6.189	≤ 6.35	≤ 9.236 (L&T design ABCD towers) ≤ 10.64 (TSP Towers)	≤ 9.235
8	Maximum operating temperature	≥180° C	≥180° C	≥180° C	≥180° C
9	Tension at 32 ° C and NIL Wind	Not exceeding 25% of UTS of proposed conductor			
10	Tension at 32 ° C and Full Wind in kgs	≤ 4610	≤ 5162	≤ 7700	≤ 8700
11	Ruling span in meters	300	320	350 (L&T design ABCD towers) 380 (TSP Towers)	350 m
12	Max. Conductor DC resistance at 20 °C in ohms/km	≤ 0.1570	≤ 0.1390	≤ 0.0686	≤ 0.05552
13	Max. Wind pressure (Zone-5) in Kg/m^2	167	167	170	172
14	Operating Voltage Level	132kV	132kV	220kV	220kV
	*The ambient temperature may be considered as 45 ° C for all ampacity				
	**Shall be equal to or higher than the respective ACSR Panther / Lynx / Zebra / Moose conductor as applicable				
	***Bidder must indicate weight of aluminium and steel per kilometre of conductor to arrive the PV Formula.				

Keeping the above parameters into consideration, **APTRANSCO may evaluate bids considering physical properties of the conductor and its compatibility to be ensured to use for APTransco transmission lines.**

The bidder shall indicate the technical particulars and details of the construction of the conductor in the relevant schedule of GTP. The bidder shall also guarantee the DC resistance of conductor at 20 deg C and AC resistance at the calculated temperature corresponding to 50Hz Ampacity in Sl.No.6 of above table and as per current corresponds to per sub conductor at specified ambient conditions (designed temperature).

The bidder has to specify UTS of core & maximum permissible load allowed on core for various sag-tension conditions at knee point. The bidder must give detailed calculation for core UTS.

The Supplier shall submit the supporting calculations for the AC resistance at Ampacity in Sl.No.6 of above table for the offered ACSS type conductor based on Cl.No.1.2.1 during detailed engineering.

NOTE: "The Guaranteed Technical Particulars for the Conductor being supplied shall be provided with the Bid as specified in the Technical Specification. The Bids without the Guaranteed Technical Particulars shall be treated as Non-Responsive."

1.4 PLS Cadd cable file

1.4.1 The bidder should submit PLS Cadd cable data (*.wir* file) for the offered ACSS conductor size (Panther/Lynx/Zebra/Moose as applicable) as per the following:

(i) Final values of modulus of elasticity of Aluminium/ Aluminium Alloy/ Core, Coefficient of Linear Expansion of Aluminium/ Aluminium Alloy/ Core, Stress-Strain coefficients & Creep coefficients of aluminium/ aluminium alloy / core in the cable data (*.wir* file) used for calculation of sag in PLSCAD shall be based on either of the following :-

a) Existing '.wir' files for offered conductor as available on PLS website.

b) A file derived from existing standard file for conductor of equivalent/ near equivalent stranding.

c) A file derived from type test conducted on conductor of same stranding.

In each of the above cases, proper justification in the form of test reports/ calculations/ print out of '.wir' file as available on PLS website, etc. shall be required to be submitted during detailed engineering.

1.4.2 Various conductor parameters (viz. modulus of elasticity, coefficient of linear expansion, stress-strain and creep, etc.) considered above in the calculation shall be verified during detailed engineering based on type tests conducted.

1.5 Workmanship

1.5.1 All the conductor strands shall be smooth, uniform and free from all imperfections, such as spills and splits, cracks, die marks, scratches, abrasions, rust etc.

1.5.2 The finished conductor shall be smooth, compact, uniform and free from all imperfections including kinks (protrusion of wires), wire cross over, over riding, looseness (wire being dislocated by finger/hand pressure and/or unusual bangle noise on tapping), material inclusions, white rust, powder formation or black spot (on account of reaction with trapped rain water etc.), dirt, grit etc.

1.6 Joints in Wires

1.6.1 Aluminium Alloy Wires

1.6.1.1 During stranding, no Aluminium/ Aluminium Alloy wire welds shall be made for the purpose of achieving the required conductor length.

1.6.1.2 **No joints** shall be permitted in the individual wires in the outer most layer of the finished conductor. However, joints are permitted in the inner layer(s) of the conductor unavoidably broken during stranding provided such breaks are not associated with either inherently defective wire or with the use of short lengths of Aluminium/ Aluminium Alloy wires. Such joints shall not be more than four (4) per conductor length and shall not be closer than 15 meters from joint in the same wire or in any other Aluminium Alloy wire of the completed conductor. A record of such joints for each individual length of the conductor shall be maintained by the Supplier for Purchaser's review.

1.6.1.3 Joints shall be made by cold pressure butt welding and shall withstand a stress of not less than the breaking strength of individual strand guaranteed.

1.6.2 Core Wires

There shall be no **joint** of any kind in the finished wire entering into the manufacture of the strand. There shall also be no joints or splices in any length of the completed core.

1.7 Tolerances

Manufacturing tolerances on the dimensions to the extent of one percent (+/-1 %) shall be permitted for individual strands and the complete conductor. In case manufacturing tolerances more than one percent (+/-1 %) is proposed for individual strands of a particular conductor as per applicable Indian/ International standard, the same may be permitted subject to submission of supporting documents/ details by the Supplier while approval of GTP during detailed engineering.

1.8 Materials

The materials used for construction of the conductor shall be such that the conductor meets the specified technical and performance requirements.

1.8.1 Outer layer

The material of outer layer of HTLS conductor shall be of annealed aluminium (0 tempered) having purity not less than 99.5% and a copper content not exceeding 0.04%. The strands shall be manufactured through appropriate manufacturing process to ensure consistent electrical, mechanical and metallurgical properties under continuous high temperature operation. Bidder shall guarantee the chemical composition in the schedule GTP of BPS and also furnish description of the manufacturing process in the Bid.

1.8.1.1 **The fully annealed type (0 tempered) aluminium strands /round wire shall be accepted.**

1.8.2 Core

1.8.2.1 The core wire strand(s) shall be of galvanized steel or Zinc-5% Aluminium – Misch metal alloy coated steel wires or aluminium clad steel wires and shall have properties conforming to the technical performance requirements of the finished conductor. In case, the designed maximum temperature of the offered HTLS conductor exceeds 180 deg C, ordinary zinc coating/ galvanizing of the Steel core wires shall not be accepted and only aluminium clad or Misch metal coated wires shall be permitted. Bidder shall furnish properties and composition of the core wire strand(s) in the schedule GTP .

1.8.2.2 The zinc used for galvanizing of core (if used) shall be electrolytic High Grade Zinc of 99.95% purity. It shall conform to and satisfy all the requirements of IS: 209. The minimum mass of zinc coating shall conform to the requirements of relevant standard as per IEC-888. Zinc-5% Aluminium –Mischmetal alloy coating, if used, shall conform to and satisfy all the requirements of ASTM B 803 / B 958.

1.8.2.3 The aluminium cladding of wires shall be with aluminium having purity not less than 99.5 % and shall be thoroughly bonded to the core wire strand(s). Shall conform to and satisfy all the requirements of IEC 1232.

1.9 Standard Length

1.9.1 The standard length of the conductor shall be indicated in the guaranteed technical particulars of offer. A tolerance of +/-5% on the standard length offered by the Bidder shall be permitted. Standard Length shall not be more than **2500** meters. All lengths outside this limit of tolerance shall be treated as random lengths.

1.9.2 Random lengths will be accepted provided no length is less than 70% of the standard length and the total quantity of such random lengths shall not be more than 10% of the total quantity ordered. At no point, the cumulative quantity supplied of such random lengths shall be more

than 12.5% of the total cumulative quantity supplied including such random lengths. However, the last 20% of the quantity ordered shall be supplied only in standard lengths as specified.

1.9.3 Bidder shall also indicate the maximum single length, above the standard length, he can manufacture in the guaranteed technical particulars of offer. This is required for special stretches like river crossing etc. The Purchaser reserves the right to place orders for the above lengths on the same terms and conditions applicable for the standard lengths during the pendency of the Contract.

2.0 Tests and Standards

2.1 Type Tests

2.1.1 Type Tests on Stranded Conductor/ Stranded wire

The following tests shall be conducted once on sample/samples of conductor from each manufacturing facility:

The following tests should have been conducted in last Ten years for which offer is made and should be submitted along with the bid. The bids without valid type test reports are liable for rejection. Type test of offered technology (for the respective ACSS equivalent conductor or higher size) will be acceptable. If any test is not conducted, the same will have to be conducted by contractor at no extra cost to APTransco. Further Type Test should have been conducted on the samples of conductor manufactured from same stranding machine by which conductor to be supplied will be manufactured after award of contract.

All the specified type tests on HTLS Conductor offered by the bidder shall not be required to be carried out if valid test certificate is available i.e., tests conducted within last seven years from the date of bid opening in an accredited laboratory or witnessed by the representative (s) of a Utility. In the event of any discrepancy in the test report (i.e., any test report not applicable due to any design/ material/manufacturing process change including substitution of components or due to non-compliance with the requirement stipulated in the Technical Specification) the tests shall be conducted by the Contractor at no extra cost to APTransco.

(i) On complete Conductor

a)	DC resistance test on stranded conductor	: As per Annexure-A
b)	UTS test on stranded conductor i) Room Temperature and ii) Elevated Temperature	: As per Annexure-A

c)	Radio interference voltage test (dry) (applicable for 400KV & above voltage rating)	: As per Annexure-A
d)	Corona extinction voltage test (dry) (applicable for 400KV & above voltage rating)	: As per Annexure-A
e)	Stress- Strain test on stranded conductor and core at room temperature	: IEC 1089
f)	Stress-strain test on stranded conductor and core at elevated temperature	As per Annexure-A
g)	High temperature endurance & creep test on stranded conductor	: As per Annexure-A
h)	Sheaves Test	: As per Annexure-A
i)	Axial Impact Test	: As per Annexure-A
j)	Radial Crush Test	: As per Annexure-A
k)	Torsional Ductility Test	: As per Annexure-A
l)	Aeolian Vibration Test	: As per Annexure-A
m)	Temperature Cycle Test	: As per Annexure-A
n)	Salt Spray Test (applicable for stranded composite core conductors having galvanic protection layer not conforming to ASTM B987)	: As per Annexure-A

(ii) On Conductor Strand/core

a)	Heat resistance test on Aluminium Alloy strands and Core	: As per Annexure-A
b)	Bending test on aluminium clad core (if applicable)	: As per Annexure-A
c)	Compression test on aluminium clad core (if applicable)	: As per Annexure-A
d)	Coefficient of linear expansion on core/ core wire	: As per Annexure-A
e)	Brittle fracture test on composite core (if applicable)	: As per Annexure-A
f)	Bending test on composite core (if applicable)	: As per Annexure-A

2.1.2 Type tests specified under Clause 2.1.1 shall not be required to be carried out if a valid test certificate is available for the offered/ design, i.e., tests conducted earlier should have been conducted in accredited laboratory (accredited based on ISO/IEC guide 25/17025 or EN 45001 by the National Accreditation body of the country where laboratory is located) or witnessed by the representative (s) of POWERGRID or any other Utility.

In the event of any discrepancy in the test report (i.e., any test report not applicable due to any design / material/manufacturing process change including substitution of components or due to non-compliance with the requirement stipulated in the Technical Specification) the tests shall be conducted by the Supplier at no extra cost to the Employer/ Purchaser.

2.2 <u>Acceptance Tests:</u>

a)	Visual and dimensional check on drum	: As per Annexure-A
b)	Visual check for joints scratches etc. and length measurement of conductor by rewinding	: As per Annexure-A
c)	Dimensional check on core strands and Aluminium/ Aluminium Alloy strands	: As per Annexure-A
d)	Check for lay-ratios of various layers	: As per Annexure-A
e)	Galvanizing test on core strands (if applicable)	: As per Annexure-A
f)	Coating Test on Zinc-5% Al-Mischmetal alloy coating (if applicable)	: As per ASTM B803/B958
g)	Adherence of Coating Test on Zinc-5% Al- Mischmetal alloy coating (if applicable)	: As per ASTM B803/B958
h)	Thickness of aluminum on aluminium clad wires	: As per Annexure-A
i)	Torsion and Elongation tests on core strands	: As per Annexure-A
j)	Breaking load test on core strands and Aluminium/ Aluminium Alloy strands	: As per Annexure-A
k)	Wrap test on core strands (if applicable) and Aluminium/ Aluminium Alloy strands	: As per IEC:888 & IEC:889
l)	Minimum conductivity test on thermal resistant Aluminium Alloy/ Aluminium strands	: As per IEC : 889
m)	Procedure qualification test on welded joint of Aluminium/ Aluminium Alloy strands	: As per Annexure-A
n)	Heat resistance test on Aluminium Alloy strands	: As per Annexure-A

o)	Ageing test on filler (if applicable)	: As per Annexure-A
p)	Minimum conductivity test on aluminium clad core wires (if applicable)	: As per Annexure-A
q)	Glass transition temperature test on composite core (if applicable)	: As per Annexure-A
r)	Flexural Strength test on composite core (if applicable)	: As per Annexure-A
s)	Bending test on composite Core (if applicable)	: As per Annexure-A
t)	Galvanic Protection Barrier Layer Thickness test on composite core (applicable for composite core conductors having galvanic barrier layer conforming to ASTM B987)	: As per ASTM B987
u)	Measurement of Ampacity at Room temperature and elevated Temperatures as per APTransco GTP.	: APTRANSCO GTP

Note: All the above tests except (m) shall be carried out on Aluminium Alloy and core strands after stranding only. Sl.No. (u) & (v) tests shall be carried out at in-house or reputed 3rd party NABL Accredited lab like CPRI/ERDA/Tag etc.

2.3 Routine Test

a) Check to ensure that the joints are as per Specification

b) Check that there are no cuts, fins etc., on the strands.

c) Check that drums are as per Specification

d) All acceptance tests as mentioned above to be carried out on 10 % of drums

2.4 Tests During Manufacture:

a)	Chemical analysis of zinc used for galvanizing	**As per Annexure-A**
b)	Chemical analysis of Aluminium alloy usedfor making Aluminium Alloy strands	**As per Annexure-A**
c)	Chemical analysis of core strands (not applicable on composite core)	**As per Annexure-A**
d)	Procedure qualification tests on welded joints or individual aluminium strands.	**As per Annexure-A**

2.5 Testing Expenses

2.5.1 A s indicated in Clause 2.1.2, no type test charges shall be payable to the Supplier.

2.5.2 In case of failure in any type test the Supplier is either required to manufacture fresh sample lot and repeat the entire test successfully once or repeat that particular type test three times successfully on the sample selected from the already manufactured lot at his own expenses. In case a fresh lot is manufactured for testing then the lot already manufactured shall be rejected.

2.5.3 Supplier shall indicate the laboratories in which they propose to conduct the type tests, if required. They shall ensure that adequate facilities are available in the laboratories and the tests can be completed in these laboratories within the time schedule guaranteed by them.

2.5.4 The entire cost of testing for the acceptance and routine tests and Tests during manufacture specified herein shall be treated as included in the quoted unit price of conductor, except for the expenses of the inspector/ Purchaser's representative.

2.5.5 In case of failure in any type test, if repeat type tests are required to be conducted, then all the expenses for deputation of Inspector/ Purchaser's representative shall be deducted from the contract price. Also if on receipt of the Supplier's notice of testing, the Purchaser's representative does not find material/ testing facilities to be ready for testing the expenses incurred by the Purchaser for re-deputation shall be deducted from contract price.

2.5.6 The Supplier shall intimate the Purchaser about carrying out of the type tests along with detailed testing programme at least 3 weeks in advance (in case of testing in India) and at least 6 weeks in advance (in case of testing abroad) of the schedule date of testing during which the Purchaser will arrange to depute his representative to be present at the time of carrying out the tests.

2.6 Additional Tests

2.6.1 The Purchaser reserves the right of having at his own expenses any other test(s) of reasonable nature carried out at Supplier's premises, at site or in any other place in addition to the aforesaid type, acceptance and routine tests to satisfy himself that the materials comply with the Specifications.

2.6.2 The Purchaser also reserves the right to conduct all the tests mentioned in this specification at his own expense on the samples drawn from the site at Supplier's premises or at any other test centre. In case of evidence of non-compliance, it shall be binding on the part of Supplier to prove the compliance of the items to the technical specifications by repeat tests, or correction of deficiencies, or replacement of defective items all without any extra cost to the Purchaser.

2.7 Sample Batch for Type Testing

2.7.1 The Supplier shall offer material for selection of samples for type testing only after getting Quality Assurance Plan approved from Purchaser's Quality Assurance Dept. The sample shall be manufactured strictly in accordance with the Quality Assurance Plan approved by Purchaser.

2.7.2 The Supplier shall offer at least three drums for selection of sample required for conducting all the type test.

2.7.3 The Supplier is required to carry out all the acceptance tests successfully in presence of Purchaser's representative before sample selection.

2.8 Test Reports

2.8.1 Copies of type test reports shall be furnished in at least three copies along with one original. One copy will be returned duly certified by the Purchaser only after which the commercial production of the material shall start.

2.8.2 Record of routine test reports shall be maintained by the Supplier at his works for periodic inspection by the Purchaser's representative.

2.8.3 Test Certificates of tests during manufacture shall be maintained by the Supplier. These shall be produced for verification as and when desired by the Purchaser.

2.9 Inspection

2.9.1 The Purchaser's representative shall at all times be entitled to have access to the works and all places of manufacture, where conductor shall be manufactured and representative shall have full facilities for unrestricted inspection of the Supplier's works, raw materials and process of manufacture for conducting necessary tests as detailed herein.

2.9.2 The Supplier shall keep the Purchaser informed in advance of the time of starting and of the progress of manufacture of conductor in its various stages so that arrangements can be made for inspection.

2.9.3 No material shall be dispatched from its point of manufacture before it has been satisfactorily inspected and tested, unless the inspection is waived off by the Purchaser in writing. In the latter case also the conductor shall be dispatched only after satisfactory testing for all tests specified herein have been completed.

2.9.4 The acceptance of any quantity of material shall in no way relieve the Supplier of any of his responsibilities for meeting all requirements of the Specification, and shall not prevent subsequent rejection if such material is later found to be defective.

2.10 Test Facilities

2.10.1 The following additional test facilities shall be available at the Supplier works:

a) Calibration of various testing and measuring equipment including tensile testing machine, resistance measurement facilities, burette, thermometer, barometer etc.

b) Standard resistance for calibration of resistance bridges.

c) Finished conductor shall be checked for length verification and surface finish on separate rewinding machine at reduced speed (variable from 8 to 16 meters per minute). The rewinding facilities shall have appropriate clutch system and free of vibrations, jerks etc. with traverse laying facilities.

2.11 Packing

2.11.1 The conductor shall be supplied in non-returnable, strong, wooden drums provided with lagging of adequate strength, constructed to protect the conductor against all damage and displacement during transit, storage and subsequent handling and stringing operations in the field. The Supplier shall be responsible for any loss or damage during transportation handling and storage due to improper packing. The drums shall generally conform to IS:1778, except as otherwise specified hereinafter.

2.11.2 The drums shall be suitable for wheel mounting and for letting off the conductor under a minimum controlled tension of the order of 5 KN.

2.11.3 The Bidder should submit their proposed drum drawings along with the bid.

2.11.4 For conductor, one standard length shall be wound on each drum.

2.11.5 The conductor ends shall be properly sealed and secured on the side of one of the flanges to avoid loosening of the conductor layers during transit and handling.

2.11.6 All wooden components shall be manufactured out of seasoned soft wood free from defects that may materially weaken the component parts of the drums. Preservative treatment shall be applied to the entire drum with preservatives of a quality which is not harmful to the conductor.

2.11.7 The flanges shall be of two ply construction with each ply at right angles to the adjacent ply and nailed together. The nails shall be driven from the inside face flange, punched and then clenched on the outer face. The thickness of each ply shall not vary by more than 3mm from that indicated in the figure. There shall be at least 3 nails per plank of ply with maximum nail spacing of 75mm. Where a slot is cut in the flange to receive the inner end of the conductor the entrance shall be in line with the periphery of the barrel.

2.11.8 The wooden battens used for making the barrel of the conductor shall be of segmental type. These shall be nailed to the barrel supports with at least two nails. The battens shall be closely butted and shall provide a round barrel with smooth external surface. The edges of the battens shall be rounded or chamfered to avoid damage to the conductor.

2.11.9 Barrel studs shall be used for the construction of drums. The flanges shall be holed and the barrel supports slotted to receive them. The barrel studs shall be threaded over a length on either end, sufficient to accommodate washers, spindle plates and nuts for fixing flanges at the required spacing.

2.11.10 Normally, the nuts on the studs shall stand protruded of the flanges. All the nails used on the inner surface of the flanges and the drum barrel shall be counter sunk. The ends of barrel shall generally be flushed with the top of the nuts.

2.11.11 The inner cheek of the flanges and drum barrel surface shall be painted with a bitumen based paint.

2.11.12 Before reeling, card board or double corrugated or thick bituminized water- proof bamboo paper shall be secured to the drum barrel and inside of flanges of the drum by means of a suitable commercial adhesive material. After reeling the conductor, the exposed surface of the outer layer of conductor shall be wrapped with water proof thick bituminized bamboo paper to preserve the conductor from dirt, grit and damage during transport and handling.

2.11.13 A minimum space of 75 mm for conductor shall be provided between the inner surface of the external protective tagging and outer layer of the conductor.

2.11.14 Each batten shall be securely nailed across grains as far as possible to the flange, edges with at least 2 nails per end. The length of the nails shall not be less than twice the thickness of the battens. The nails shall not protrude above the general surface and shall not have exposed sharp, edges or allow the battens to be released due to corrosion.

2.11.15 The nuts on the barrel studs shall be tack welded on the one side in order to fully secure them. On the second end, a spring washer shall be used.

2.11.16 A steel collar shall be used to secure all barrel studs. This collar shall be located between the washers and the steal drum and secured to the central steel plate by welding.

2.11.17 Outside the protective lagging, there shall be minimum of two binder consisting of hoop iron/galvanized steel wire. Each protective lagging shall have two recesses to accommodate the binders.

2.11.18 As an alternative to wooden drum, Supplier may also supply the conductors in returnable/ non-returnable painted steel drums. After preparation of steel surface according to IS:9954, synthetic enamel paint shall be applied after application of one coat of primer. Wooden/Steel drum will be treated at par for evaluation purpose and accordingly the Bidder should quote in the package.

2.11.19 In case of returnable steel drums for conductor, following clauses shall apply:-

a) The Ownership of the empty conductor drums shall lie with the conductor Supplier who shall ultimately take back the empty conductor drum from the Project site(s) from the erection Supplier's designated stores after the running out of conductor from the drum.

b) The erection contractor shall intimate the Conductor supplier and Purchaser regarding empty steel drums at their designated stores. Necessary coordination for taking back the empty steel drums in this regard shall be done by the Conductor Supplier with the erection Contractor.

c) The empty drum shall be taken back by the conductor supplier from the stores of erection contractor as & when these are available after usage of conductor. Conductor Supplier shall be required to take back the empty steel drum within a period of one month from date of information by erection contractor regarding availability of the drums at erection contractor stores. However, 2% of the total drums shall not be returned to the conductor Supplier as these may be used for storage of spare conductor by the Purchaser. The steel drums may get damage and wear & tear due to transportation, normal handling & operation at site, which shall be rectified by the conductor Supplier before re-use. However, 2% of the total drums shall not be returned on account of damages / wastage for which no compensation will be payable. The wastage beyond 2% shall be reimbursed by Erection Contractor.

Thus, 4% of total drums shall not be returnable to the conductor Supplier.

2.11.20 As an alternative to outer wooden lagging, in case of returnable/ non-returnable steel drums, solid polypropylene sheet (of min 5mm thickness) can be used for outer covering of conductor. In case of PP sheets are proposed to be used by the Supplier, the conductor Supplier shall supply two nos additional binders per drum for re-wrapping PP sheet with each lot of conductor and 5 nos. crimping machines with the first lot of conductor for crimping the binders at site.

2.11.21 **Marking**

Each drum shall have the following information stenciled on it in indelible ink along with other essential data :

(a) Contract/Award letter number.

(b) Name and address of consignee.

(c) Manufacturer's name and address.

(d) Drum number

(e) Size of conductor

(f) Length of conductor in meters

(g) Arrow marking for unwinding

(h) Position of the conductor ends

(i) Distance between outer-most Layer of conductor and the inner surface of lagging.

(k) Barrel diameter at three locations & an arrow marking at the location of the measurement.

(l) Number of turns in the outer most layer.

(m) Gross weight of drum after putting lagging. (n)

Tear weight of the drum without lagging.

(o) Net weight of the conductor in the drum.

(p) CIP/MICC No.

The above should be indicated in the packing list also.

2.12 Verification of Conductor Length

The Purchaser reserves the right to verify the length of conductor after unreeling at least ten (10) percent of the drums in a lot offered for inspection.

2.13 Standards

2.13.1 The conductor shall conform to the following Indian/International Standards, which shall mean latest revisions, with amendments/changes adopted and published, unless specifically stated otherwise in the Specification.

2.13.2 In the event of the supply of conductor conforming to standards other than specified, the Bidder shall confirm in his bid that these standards are equivalent to those specified. In case of award, salient features of comparison between the standards proposed by the Supplier and those specified in this document will be provided by the Supplier to establish their equivalence.

Sl. No.	Indian Standard	Title	International Standard
1.	IS: 209-1992	Specification for zinc	BS:3436-1986
2.	IS: 398-1992	Specification for Aluminium Conductors for Overhead Transmission Purposes	IEC:1089-1991BS:215-1970
3.	IS:398-1990 Part-II	Aluminum Conductor Galvanised Steel Reinforced	BS;215-1970 IEC:1089-1991
4.	IS : 1778-1980	Reels and Drums for Bare Conductors	BS:1559-1949
5.	IS : 1521-1991	Method of Tensile Testing of Steel Wire	ISO 6892-1984
6.	IS : 2629-1990	Recommended Practice for Hot Dip Galvanising of Iron and Steel	
7.	IS : 2633-1992	Method of Testing Uniformity of Coating on Zinc Coated Articles	
8.	IS : 4826-1992	Galvanised Coating on Round Steel Wires	IEC : 888-1987BS:443-1969
9.	IS : 6745-1990	Methods of Determination of Weight of Zinc Coating of Zinc Coated Iron and Steel Articles	BS:433-1969 ISO 1460 – 1973
10.	IS : 8263-1990	Method of Radio Interference Tests on High Voltage Insulators	IEC:437-1973 NEMA:107-1964 CISPR
11.	IS : 9997-1988	Aluminium Alloy Redraw Rods	IEC 104 – 1987
12.		Zinc Coated steel wires for stranded Conductors	IEC : 888-1987
13.		Hard drawn Aluminium wire for overhead line conductors	IEC : 889-1987
14.	IS:398 (Part-IV)	Aluminium Alloy stranded conductor	IEC : 208-1966 BS-3242-1970
15.		Aluminium clad steel wires	IEC:1232
16.		Method of measurement of resistivity of metallic materials	IEC:468
17.		Ampacity	IEEE738
18		Thermal resistant Aluminium Alloy	IEC 62004
19.		Carbon Fibre Composite core (CFCC/TS) for use in Overhead Electrical Conductors	ASTM B987

The standards mentioned above are available from:

Reference Abbreviation	Name and Address
BS	British Standards, British Standards Institution 101, Pentonvile Road, N - 19- NDUK
IEC/CISPR	International Electro technical Commission, Bureau Central de la Commission, electro Technique international, 1 Rue de verembe, Geneva SWITZERLAN D
BIS/IS	Beureau Of Indian Standards. Manak Bhavan, 9, Bahadur Shah Zafar Marg, New Delhi - 110001. INDIA
ISO	International Organisation for Standardization. Danish Board of Standardization Danish Standardizing Sraat, Aurehoegvej-12 DK-2900, Heeleprup, DENMARK.
NEMA	National Electric Manufacture Association, 155, East 44th Street. New York, NY 10017 U.S.A.
ASTM	ASTM International 100 Barr Harbor Drive,PO Box C700, West Conshohocken, PA 19428-2959 USA

Annexure-A

1. Tests on Conductor

1.1 UTS Test on Stranded Conductor at a) room temperature and b) elevated temperature

a) UTS test on stranded conductor at room temperature*

Circles perpendicular to the axis of the conductor shall be marked at two places on a sample of conductor of minimum 5 m length between fixing arrangement suitably fixed by appropriate fittings on a tensile testing machine. The load shall be increased at a steady rate up to 50% of minimum specified UTS and held for one minute. The circles drawn shall not be distorted due to relative movement of strands. Thereafter the load shall be increased at steady rate to minimum UTS and held for one minute. The Conductor sample shall not fail during this period. The applied load shall then be increased until the failing load is reached and the value recorded.

*the test is to be conducted at ambient temperature, between minimum and maximum ambient temperature of 0 deg C and 50 deg C respectively**.**

b) UTS test on stranded conductor at elevated temperature

UTS Test on Stranded Conductor shall be conducted as per clause no. 1.1(a) specified above keeping conductor temperature at the designed maximum temperature.

1.2 Corona Extinction Voltage Test (Applicable for ≥400KV lines only)

The sample assembly with each conductor of 5 m length shall be strung as per the configuration shown in the Table below:

Line Configuration	No. of conductors per Bundle	Sub-conductor Spacing (mm)	Maximum Height of the conductor above ground(m)	Minimum Corona extinction voltage (kV)
400kV with ACSS equivalent conductor to ACSR Moose	2/4	450	8.84	305

The sample assembly when subjected to power frequency/dc voltage under dry condition shall have a corona extinction voltage of not less than the values indicated in the table above. There shall be no evidence of corona on any part of the samples. The test should be conducted without corona control rings. However, small corona control rings may be used to prevent corona in the end fittings. The atmospheric conditions during testing shall be recorded and test results should be corrected for standard atmospheric conditions.

1.3 Radio Interference Voltage Test(Applicable for ≥400KV lines only)

Under the conditions as specified under (1.2) above, the conductor samples shall have radio interference voltage level below 1000 microvolts at one MHz when subjected to 50 Hz AC voltage of **305 kV line** to ground under dry conditions. This test may be carried out with corona control rings and arcing horns.

1.4 D.C. Resistance Test on Stranded Conductor

On a conductor sample of minimum 5m length two contact-clamps shall be fixed with a predetermined bolt torque. The resistance shall be measured by a Kelvin double bridge or using micro ohm meter of suitable accuracy by placing the clamps initially zero metre and subsequently one metre apart. The test shall be repeated at least five times and the average value recorded. The value obtained shall be corrected to the value at 20deg C as per IS:398-(Part-IV)/(Part-V). The resistance corrected at 20deg C shall conform to the requirements of this Specification.

1.5 Stress-strain test at elevated temperature

Stress-strain test as per IEC-61089 shall be conducted keeping conductor temperature at designed maximum temperature. UTS for this test shall be 70% of the UTS guaranteed in the GTP.

1.6 High Temperature endurance & creep test

Two conductor samples of length equal to at least 100 X d + 2 X a (where, d is the conductor diameter and a is the distance between the end fitting and the gauge length) shall be strung at tension equal to 25 % of conductor UTS. The distance, a, shall be at least 25 % of the gauge length or 2 m whichever is the smaller. The conductor samples shall be subjected to tests as indicated below:

(i) On one of the conductor samples, the conductor temperature shall be maintained at 20 deg C for 1000 hours. The elongation/creep strain of the conductor during this period shall be measured and recorded at end of 1 hour, 10 hour, 100 hour and subsequently every 100 hour upto 1000 hours time period.

(ii) On other conductor sample, the conductor temperature shall be increased to designed maximum temperature in steps of 20 deg. C and thermal elongation of the conductor sample shall be measured & recorded at each step. The temperature shall be held at each step for sufficient duration for stabilization of temperature. Further, the temperature of the conductor shall be maintained at designed maximum temperature ($\pm$10 Deg. C) for 1000

hours. The elongation/creep strain of the conductor during this period shall be measured and recorded at end of 1 hour, 10 hour, 100 hour and subsequently every 100 hour upto 1000 hours time period. After completion of the above, the conductor sample shall be subjected to UTS test as mentioned above at clause 1.1 of Annexure-A. The conductor core shall withstand a load equivalent to 95 % of UTS. In case of polymer composite core conductor, the flexural strength & glass transition temperature of the core shall also be evaluated and the same shall not be degraded by more than 10 % over the value guaranteed in GTP. The value of Tg after the test, shall however, in no case be less than the design maximum temperature of conductor. The supplier shall plot the thermal elongation with temperature.

The Supplier shall furnish details of creep characteristic in respect of the conductor based on laboratory test and other laboratory investigations/ experiments conducted on similar type of conductor and shall indicate creep strain values corresponding to 1 month, 6 month, 1 year, 10 year & 20 year creep at everyday tension & designed maximum temperature as well as room temperature.

1.7 Sheaves Test

The conductor sample of minimum length of 35 meter shall be tensioned at 20 % of the UTS; pulley diameter shall be atleast 32 times that of the conductor; angle between the pulleys shall be 20 degrees. The conductor shall be passed over the pulleys 36 times a speed of 2 m/sec. After this test UTS test on the conductor shall be carried out as mentioned above at clause 1.1 of Annexure-A. In the case of polymer composite core conductors, the core shall be inspected for damage by subjecting the core to a dye penetration test as per ASTM B987 section 14. Dye penetrate exposure time shall be 30 +1/-0 minutes.

1.8 Axial Impact Test

The conductor sample shall be suspended vertically and load applied by dropping a 650 Kg from an elevation of 4 meters above the sample. The impact velocity shall be not be less than 8 m/sec. with an initial pre-tension of 200 kgs. The curve for load vs time shall be recorded and recorded load of failure for core shall not be less than UTS of core.

1.9 Radial Crush Test

A section of conductor is to be crushed between two six inch steel platens. Load shall be held at 350 Kgs for 1 minute and then released. Core strands shall be subsequently disassembled and tensile tested. Core strands shall exhibit full strength retention vis-a-vis guaranteed breaking strength of core wires (after stranding).

1.10 Torsional Ductility Test

The conductor sample of 10-15 meters shall be loaded to 20% of UTS and then rotated in increasing steps of +/-180 deg. The entire conductor shall withstand atleast 16 such rotation and there shall not be any damage to Aluminium/ Aluminium Alloy or core wires.

In case of composite core conductors, after 4 rotations or after separation of aluminium strands, the aluminium wires shall be cut and removed from the conductor and the exposed core shall be twisted and shall withstand upto 16 rotations.

1.11 Aeolian Vibration Test

The conductor and supporting hardware shall be loaded to 25% of UTS. A dynamometer, load cell, calibrated beam or other device shall be used to measure the conductor tension. Some means should be provided to maintain constant tension to allow for temperature fluctuations during the testing. The overall span between system terminations shall be a minimum of 30 m. The span shall be supported at a height such that the static sag angle of the cable to horizontal is (1.5 $\pm$ 0.5) deg in the active span. Means shall be provided for measuring and monitoring the mid-loop (antinode) vibration amplitude at a free loop, not a support loop. An electronically controlled shaker shall be used to excite the conductor in the vertical plane. The shaker armature shall be securely fastened to the conductor so it is perpendicular to the conductor in the vertical plane. The shaker should be located in the span to allow for a minimum of six vibration loops between the suspension assembly and the shaker

The test shall be carried out at one or more resonance frequencies (more than 10 Hz) . The amplitude (peak-to-peak) at the antinode point shall be one third of conductor diameter. The assembly shall be vibrated for not less than 10 million cycles without any failure. After the test, the conductor should not exhibit any damage (broken strands). The conductor shall be tested to demonstrate that it retains at least 95% UTS.

1.12 Temperature Cycle Test

The purpose of this test is verification of degradation characteristics of metallic and non-metallic material when subjected to thermal cycling temperature cycling can create large internal stresses due to thermal expansion mismatch between constituents**.**

Test Methods:-

- Mechanical tension, 20 % UTS, marks on the conductor at the edge of the conductor

- 100 cycles from room temperature up to designed maximum operating temperature.

  Hold at designed maximum temperature ± 2.5 deg. C for 5 minutes.
- After the above mentioned 100 cycle, the Mechanical tension shall be increased up to 70 % UTS at room temperature and kept at this tension for 24 hours. Thereafter, release to 20% UTS
- This cycling test shall be repeated 5 times.
- During the test, temperature of connectors, conductor and resistance are recorded according to ANSI C 119.
- A breaking load test is applied at the end of the test. Conductor strength has to be higher than 95 % UTS.
- In case of polymer composites, after thermal cycling, the flexural strength and the Glass Transition temperature (Tg) shall not degrade by more than10 % of the value guaranteed in GTP. Flexural strength shall be obtained on the basis of test procedure indicated at 1.32 below. The value of Tg after the test, shall however, in no case be less than the design maximum temperature of conductor.

1.13 Heat Resistance test on Aluminium Alloy wire

Breaking load test as per clause 1.25 below shall be carried out before and after heating the sample in uniform heat furnace at following temperature for one hour. The breaking strength of the wire after heating shall not be less than the 90% of the breaking strength before heating:-

Maximum continuous operating temperature of the conductor	Test Temperature
Upto 150 deg. C	230 degC (+5/-3 degC)
More than 150 deg. C & upto 210 deg. C	280 degC (+5/-3 degC)
More than 210 deg. C & upto 230 deg. C	400 degC (+5/-3 degC)

1.14 Bending test on aluminium clad core wire (if applicable)

A sample of aluminium clad invar strand measuring 30 cm in length shall be subject to bending with help of a vise. The vised length of wire should be 5 cm and radius of bend 4.8 mm. The bending should be first 90 degrees left and 90 degree right. After this operation the strand should cut at the bending point. There should be no separation of core and aluminium at the bending point after this operation.

1.15 Compression test on aluminium clad wires

A sample of aluminium clad core strand 10 mm in length is to be compressed by a plate with a load of 3600 kgs. The aluminium clad core strand should not break.

1.16 Coefficient of linear expansion for core wires

The temperature and elongation on a sample shall be continuously measured and recorded at interval of approximately 15 degree C from 15 degree C to designed maximum temperature corresponding to rated current as per requirement, by changing the temperature by suitable means. Coefficient of linear expansion shall be determined from the measured results.

1.17 Brittle fracture test for polymer composite core only (if applicable only)

The core sample shall be tensioned to approx. 25 % of UTS with simultaneous application of 1N-HNO3 acid directly in contact with naked polymer composite core for 96 hrs. The contact length of acid shall not be less than 40mm and thickness around the core not less than 10mm. The core shall withstand UTS test after 96 hours.

1.18 Visual and Dimensional Check on Drums

The drums shall be visually and dimensionally checked to ensure that they conform to the approved drawings.

1.19 Visual Check for Joints, Scratches etc.

Conductor drums shall be rewound in the presence of the Purchaser. The Purchaser shall visually check for scratches, joints etc. and that the conductor generally conform to the requirements of this Specification. Ten percent (10%) drums from each lot shall be rewound in the presence of the Purchaser's representative.

1.20 Dimensional Check on Core Wires and Aluminium/ Aluminium Alloy Wires

The individual strands shall be dimensionally checked to ensure that they conform to the requirement of this Specification. Diameter of formed wires shall be determined as per clause 6.3.1.2 of IEC 62420.

1.21 Check for Lay-ratios of Various Layers

The lay-ratios of various layers shall be checked to ensure that they conform to the guaranteed values furnished by the purchaser.

1.22 Galvanising Test

The test procedure shall be as specified in IEC: 888. The material shall conform to the requirements of this Specification. The adherence of zinc shall be checked by wrapping around a mandrel four times the diameter of steel wire.

1.23 Aluminum thickness on aluminum clad wires (if applicable)

The thickness of aluminium of the specimen shall be determined by using suitable electrical indicating instruments operating on the permeameter principle, or direct measurement. Measurements shall be read to three decimal places, and number rounded to two decimal places is considered as measured thickness. For reference purposes, direct measurement shall be used to determine aluminium thickness on specimens taken from the end of the coils.

1.24 Torsion and Elongation Tests on Core wire/ Composite core

The test procedures for Torsion and Elongation Tests on Core wires shall beas per clause No. 6.3.3 and 6.3.2 b) of IEC 61232 respectively. In torsion test, the number of complete twists before fracture shall not be less than the value specified in the GTP on a length equal to 100 times the standard diameter of the strand. In case test sample length is less or more than 100 times the stranded diameter of the strand, the minimum number of twists will be proportioned to the length and if number comes in the fraction then it will be rounded off to next higher whole number. In elongation test, the elongation at fracture of the strand shall not be less than the value specified in the GTP for a gauge length of 250 mm.

1.25 Breaking load test on Aluminium/ Aluminium Alloy & Core wires/composite core and D.C Resistance test on Aluminium/ Aluminium Alloy wire

The above tests shall be carried out as per IEC: 888/889 and the results shall meet the requirements of the specification.

1.26 Wrap test on Core wires

The wrap test on core strands shall be meet the requirements of IEC: 888. In case of aluminium clad core wire, the same shall be wrapped around a mandel of diameter of five times that of the strand to form a helix of eight turns. The strand shall be unwrapped. No breakage of strand shall occurred.

1.27 Minimum conductivity test on thermal resistant aluminium alloy wire:

Resistivity test as per IEC-468/IEC 889 shall be conducted to confirm minimum conductivity as per specification requirement.

1.28 Procedure Qualification test on welded Aluminium/ Aluminium Alloy wire.

Two Aluminium Alloy wire shall be welded as per the approved quality plan and shall be subjected to tensile load. The breaking strength of the welded joint of the wire shall not be less than the guaranteed breaking strength of individual strands.

1.29 Ageing Test on Filler (if applicable)

The test shall be done in accordance with Grease drop point test method. The specimen should be drop as a droplet when kept at a temperature 40 deg. C above designed maximum temperature of the conductor for 30 minutes. The temperature shall then be increase till one droplet drops and the temperature recorded.

1.30 **Aluminium conductivity test on aluminium clad wire (if applicable)**
Resistivity test as per IEC-468 shall be conducted to confirm minimum conductivity as per specification requirement.

1.31 **Glass Transition Temperature Test on composite core only (if required)**
Test shall be conducted as per ASTM B987. The minimum glass transition temperature shall be either (i) the design maximum continuous operating temperature of the offered HTLS conductor + 35 Deg C or (ii) minimum glass transition temperature as per ASTM B987 i.e.180 deg. C + 25 Deg C ; Whichever is lower. In case, the design maximum continuous operating temperature of the offered HTLS conductor is more than the minimum glass transition temperature as per ASTM B987 i.e. more than 180 deg. C, the test shall be conducted as per ASTM B987 & the minimum glass transition temperature shall be the design maximum continuous operating temperature of the offered HTLS conductor + 25 Deg C.

1.32 **Flexural Strength Test for composite core (if applicable only)**

Supplier can carry out the test as per ASTM D7264, ASTM D4475 or ISO 14125. The flexural strength shall not be less than the value guaranteed in the GTP.

1.33 **Bending Test on Polymer Composite Core (Type Test) (if applicable only)**

Bending test on polymer composite core (CFC) before stranding shall be performed as per ASTM B987/B987M-17 on polymer composite core samples taken from composite core at conductor manufacturing unit before stranding of conductor. Alternatively, supplier may carry out bending test on polymer composite core (CFC) before stranding on the samples taken at the core manufacturing unit, from the same reel being supplied to conductor manufacturer subject to proper traceability of the same at the conductor manufacturers works.

Bending test on polymer composite core (CFC) shall also be performed as per ASTM B987/B987M-17 on polymer composite core samples taken from stranded conductor. For test after stranding the diameter of cylindrical mandrel shall be as following:

For high strength grade CFC – 60 times the diameter of CFC

For Extra high strength grade CFC – 70 times the diameter of CFC

1.34 **Bending test on polymer composite core (Acceptance test): (if applicable only)**
Bending test on polymer composite core (CFC) shall be performed as per ASTM B987/B987M-17 on polymer composite core samples taken from stranded conductor. For test after stranding the diameter of cylindrical mandrel shall be as following:

1) For high strength grade CFC – 60 times the diameter of CFC

2) For Extra high strength grade CFC – 70 times the diameter of CFC

1.35 Salt Spray Test

The test shall be performed as per ASTM B117 on 3 samples of conductor, each 1 m in length. The three samples shall be simultaneously subjected to salt spray exposure in an environmental chamber for 1000 hours. After this test, the individual strands and composite core shall be subjected to visual examination and tensile strength test. There shall be no pitting, corrosion or deterioration on any component of the conductor after 1000 hrs of salt-spray exposure. The individual strands of aluminium/ aluminium alloy and the composite core shall retain 95% of the strength guaranteed in GTP.

1.36 Chemical Analysis of Aluminium/ Aluminium Alloy and Steel/ InvarCore Wires

Samples taken from the Aluminium/ Aluminium Alloy and core coils/strands shall be chemically/spectrographically analysed. The same shall be in conformity to the particulars guaranteed by the bidder so as to meet the requirements stated in this Specification.

1.37 Chemical Analysis of Zinc (if applicable only)

Samples taken from the zinc ingots shall be chemically/ spectrographically analysed. The same shall be in conformity to the requirements stated in the Specification.

ANNEXURE-IV

GUARANTEED TECHNICAL PARTICULARS/TECHNICAL DATA SHEET

(TO BE FILLED BY BIDDER)

TYPICAL GUARANTEED TECHNICAL PARTICULARS OF ACSS CONDUCTOR

Sl.	Description	Unit	Value guaranteed by the Bidder
1.	Name & address of Manufacturer		
2.	Construction of conductor/ Designation of conductor as per IEC:1089		
3.	**PARTICULARS OF RAW MATERIALS**		
3.1	Outer Layers a) Applicable Standard(if any) b) Type of Aluminum alloy c) Minimum purity of aluminum d) Maximum Copper content e) Zirconium content i) Maximum ii) Minimum f) Other elements---------- i) ---------- ii) ----------	 % % % % % %	
3.2	Inner Core a) Applicable Standard(if any) b) Material of core c) Chemical composition of core i) ---------- ii) ----------	 % %	
3.3	Zinc used for galvanization of inner core (if applicable) a) Minimum purity of zinc	%	
3.4	Chemical Composition of Misch Metal coating on core wires (if applicable) i) Zinc ii) Aluminium iii) Other elements----------	 % % %	
3.5	Aluminium used for Aluminium Cladding (if applicable) a) Minimum purity of aluminum b) Maximum Copper content c) Other elements---------- i) ---------- ii)……..	 % % % %	

4.	**STRANDS OF OUTER CONDUCTING PART (AFTER STRANDING)**		
4.1	Number of outer layers	Nos.	
4.2	Number of strands a) 1st Layer from core b) 2nd Layer from core c) 3rd Layer from core …… ……..	 Nos. Nos. Nos. ……. ……..	
4.2	Diameter of strands a) Nominal b) Maximum c) Minimum	 mm mm mm	
4.3	Minimum Breaking load of strand a) Before stranding b) After stranding	 kN kN	
4.4	Resistance of 1m length of strand at 20 deg. C	Ohm	
4.5	Final Modulus of elasticity	Kg/sq. mm	
4.6	Final Coefficient of linear expansion	Per ^{0} C	
5	**INNER CORE STRANDS/ INNER CORE (AFTER STRANDING)**		
5.1	Number of layers in inner core (excluding central wire)		
5.2	Number of strands a) 1st Layer from centre (excluding central wire) b) 2nd Layer from centre c) 3rd Layer from centre …… ……..	 Nos. Nos. Nos. ……. ……..	
5.3	Diameter a) Nominal b) Maximum c) Minimum	 mm mm mm	
5.3	Minimum Breaking load of strand/Core a) Before stranding b) After stranding	 kN kN	
5.4	Resistance of 1m length of strand at 20 deg. C	Ohm	

5.5	Final Modulus of elasticity	Kg/sq. mm	
5.6	Final coefficient of linear expansion	Per ^{0} C	
5.7	**Aluminum cladding of core (if applicable)**		
	a) Thickness of cladding		
	i) Maximum	mm	
	ii) Minimum	mm	
	b) Minimum no. of twists in a gauge length equal to 100 times diameter of wire which the strands can withstand in the torsion test		
	a) Before stranding	Nos.	
	b) After stranding	Nos.	
	c) Minimum elongation of strand for a gauge length of 250 mm	%	
	d) Resistance of 1m length of strand at 20 deg. C	Ohm	
5.8	**Galvanizing/ Misch Metal coating (if applicable)**		
	a) Minimum mass of zinc coating per sqm. of uncoated wire surface.	gm	
	b) Minimum mass of Misch metal coating per sqm. of uncoated wire surface (if applicable).	Nos.	
	c) Min. no. of twists which a single strand shall withstand during torsion test for a length equal to100times dia. of wire after stranding.	Nos.	
	d) Minimum elongation of strand for a gauge length of 250 mm	%	
6	**FILLER (if applicable)**		
6.1	Type & Designation of Filler		
6.2	Chemical composition of Filler		
6.3	Mass of Filler	Kg/km	
7	**COMPLETE HTLS CONDUCTOR**		
7.1	Cross section drawing of the offered conductor enclosed	Yes/No	
7.2	Diameter of conductor a) Nominal b) Maximum	 mm mm	

	c) Minimum	mm		
7.3	UTS (minimum) of Conductor	kN		
7.4	Lay ratio of conductor a) 1st layer from centre (excluding central wire) b) 2nd Layer c) 3rd Layer d) 4th Layer		Maximum	Minimum
7.5	DC resistance of conductor at 20°C	Ohm/km		
7.6	Final Modulus of elasticity			
	a) Upto transition temperature	Kg/sq. mm		
	b) Above transition temperature	Kg/sq. mm		
7.7	Coefficient of linear expansion			
	a) Upto transition temperature	Per deg C		
	b) Above transition temperature	Per deg C		
7.8	Calculation for transition temperature enclosed	Yes/No		
7.9	Transition temperature (corresponding to ---m ruling span and tension at ruling condition as per 7.19)	Deg C		
7.10	Minimum Corona Extinction Voltage (line to ground) under Dry condition[	kV(rms)		
7.11	RIV at under dry conditions	Micro-volts		
7.12	Maximum permissible conductor temperature for continuous operation	Deg C		
7.13	Maximum permissible conductor temperature for short term operation	Deg C		
7.14	Permissible duration of above short term operation	Minutes		
7.15	Steady state conductor temperature at specified conductor current of -----Amps and under Ambient conditions			

7.16	AC resistance at maximum continuous operating temperature corresponding to specified maximum operating current (---A under ambient condition	Ohm/km	
7.17	AC resistance at continuous operating temperature corresponding to specified operating current of ---A	Ohm/km	
7.18	Details of Creep characteristic for HTLS conductor enclosed	Yes/No	
7.19	**Sag Tension Calculation**		
7.19.1	Sag Tension Calculation enclosed	Yes/No	
7.19.2	Tension at 32 deg. C & no wind	Kg	
7.19.3	Sag & tension at continuous operating temperature (specified in 7.15)	Meters & Kgs	
i)	Tension for following conditions:		
a.	32 deg. C & full wind condition	kg	
b.	32 deg. C & Nil wind condition	kg	
c.	Minimum tempt. & Nil wind condition	kg	
d.	Minimum tempt. & 36% of full wind Condition	kg	
e.	32 deg. C & 75% of full wind condition	kg	
7.19.4	Tension at transition temperature	kg	
7.20	Direction of lay for outside layer		
7.21	Linear mass of the Conductor a) Standard b) Minimum c) Maximum	 Kg/km Kg/km Kg/km	
7.22	Standard length of conductor	M	

7.23	Maximum length of conductor that can be offered as single length	M	
7.24	Tolerance on standard length of conductor	%	
7.25	Drum is as per specification	Yes/No	
7.26	Nominal cross sectional area of conductor		
	a. Aluminium	Sq.mm	
	b. Steel Core (EHS/UHS/if any)	Sq.mm	
	c. Total	Sq.mm	
7.27	Nominal weight of the conductor		
	a. Aluminium	Kg/km	
	b. Steel core	Kg/km	
	c. Grease	Kg/km	
	d. Total	Kg/km	
7.28	Current carrying capacity at maximum operating temperature		
7.29	Maximum operating temperature		
7.30	Any other details which are relevant as per Indian and international standards		